* Solutions *

1. (8 pts) Fill in the blanks. No work required. No partial credit will be given. All numbers must be simplified and answers must be exact.

(1.1) Assuming that $\nabla f(2,-2) = \langle -4, 24\rangle$ and $f(2,-2) = -12$. The equation of the tangent plane to the graph of f at $(2,-2)$ is given by $z = -12 - 4(x-2) + 24(y+2)$.

OR $z = 44 - 4x + 24y$

(1.2) Assume $\nabla f(P) = \langle -3, 3\rangle$

The directional derivative of f at P in the direction of $\langle -4, 3\rangle$ is $21/5$.

$D_{\vec{u}} f(P) = \langle -3, 3\rangle \cdot \langle -4/5, 3/5\rangle = \frac{12}{5} + \frac{9}{5}$

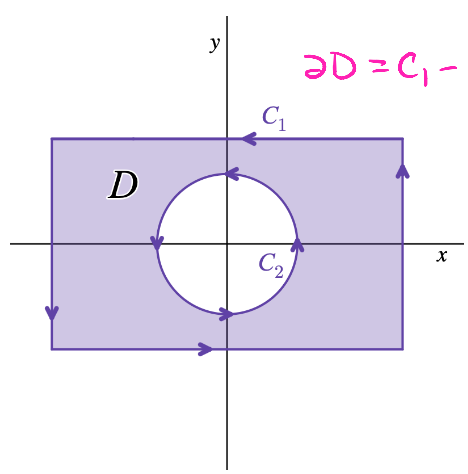
(1.3) If $f(x,y) = 2$, then $\iint_{\mathcal{D}} 2\,dA =$ $2 \cdot \text{Area}(D)$.

(1.4) If $\mathbf{F}$ is a conservative vector field and $\mathcal{C}$ is a closed curve, then $\oint_{\mathcal{C}} \mathbf{F}\cdot d\mathbf{r} =$ 0.

2. (5 pts) Consider the region below. If $\oint_{C_2} \mathbf{F}\cdot d\mathbf{r} = 12$ and $\iint_D \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}\right) dA = 48 - 4\pi$, calculate $\oint_{C_1} \mathbf{F}\cdot d\mathbf{r}$. Simplify your answer and write your final answer on line provided.

$\oint_{C_1} \mathbf{F}\cdot d\mathbf{r} =$ $60 - 4\pi$.

$\partial D = C_1 - C_2$



$$\oint_{\partial D} \vec{F}\cdot d\vec{r} = \oint_{C_1} \vec{F}\cdot d\vec{r} - \oint_{C_2} \vec{F}\cdot d\vec{r}$$
$$= \oint_{C_1} \vec{F}\cdot d\vec{r} - 12$$

$$\oint_{\partial D} \vec{F}\cdot d\vec{r} = \iint_D \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}\right) dA = 48 - 4\pi$$

$$\Rightarrow \oint_{C_1} \vec{F}\cdot d\vec{r} - 12 = 48 - 4\pi \quad \Rightarrow \oint_{C_1} \vec{F}\cdot d\vec{r} = 60 - 4\pi$$

3. For each question, credit will be given based on the selected answer(s) only. Do not make stray marks on other answers. If you need to change an answer, fully erase or make sure it is clear what answer(s) you are selecting. No work required.

(3.1) (3 pts) Assume $f_x(a,b) = f_y(a,b) = 0$. If $f_{xx}(a,b) = 2$, $f_{yy}(a,b) = 5$, and $f_{xy}(a,b) = -3$. Apply the second derivative test to classify f at (a,b). **Select <u>one</u> answer.**

- (A) f has a local maximum at (a,b).
- (B) f has a absolute maximum at (a,b).
- (C) f has a local minimum at (a,b).
- (D) f has a absolute minimum at (a,b).
- (E) f has a saddle point at (a,b).
- (F) Second derivative test is inconclusive at (a,b).

$D(a,b) = \begin{vmatrix} 2 & -3 \\ -3 & 5 \end{vmatrix} = 10 - 9 = 1 > 0$ & $f_{xx}(a,b) = 2 > 0$ ✓

(3.2) (3 pts) Let $f(x,y) = \ln(x+y)$ and $g(x,y) = \sqrt{x+y}$. **Select <u>one</u> answer.** The point $(0,0)$ is a critical point for

- (A) f but not g
- (B) neither f nor g
- (C) g but not f
- (D) both f and g

$(0,0) \notin$ Domain of f ; $(0,0) \in$ Domain of g

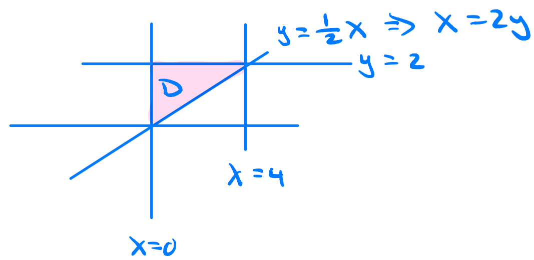
$g_x = \frac{1}{2\sqrt{x+y}}$, $g_y = \frac{1}{2\sqrt{x+y}}$

$g_x(0,0)$ DNE & $g_y(0,0)$ DNE

(3.3) (6 pts) Consider the integral $\int_0^4 \int_{\frac{1}{2}x}^2 6xe^{y^3}\,dy\,dx$. Which of the following is the integral obtained by reversing the order of integration? **Select <u>one</u> answer.**

- (A) $\int_0^2 \int_{2y}^0 6xe^{y^3}\,dx\,dy$
- (B) $\int_0^4 \int_0^{\frac{1}{2}y} 6xe^{y^3}\,dx\,dy$
- (C) $\int_0^2 \int_0^{2y} 6xe^{y^3}\,dx\,dy$
- (D) $\int_0^2 \int_{2y}^4 6xe^{y^3}\,dx\,dy$
- (E) $\int_0^2 \int_0^4 6xe^{y^3}\,dx\,dy$
- (F) $\int_{\frac{1}{2}x}^2 \int_0^4 6xe^{y^3}\,dx\,dy$
- (G) $\int_2^0 \int_{2y}^0 6xe^{y^3}\,dx\,dy$
- (H) $\int_0^4 \int_{\frac{1}{2}y}^2 6xe^{y^3}\,dx\,dy$

D: $0 \le x \le 4$, $\frac{1}{2}x \le y \le 2$ (vertically simple)



D : $0 \le x \le 2y$, $0 \le y \le 2$

(3.4) (6 pts) Consider the conservative vector field $\mathbf{F} = \langle 6xy, 3x^2 + 8y^3z^2, 4y^4z + 3z^2\rangle$. Identify a potential function for $\mathbf{F}$ if it exists. **Select <u>one</u> answer.**

- (A) $f(x,y,z) = \langle 3x^2y, 2y^4z^2, z^3\rangle$
- (B) $f(x,y,z) = 6y + 24yz^2 + 4y^4 + 6z$
- (C) $f(x,y,z) = \langle 3x^2y, 3x^2y + 2y^4z^2, 2y^4z^2 + z^3\rangle$
- (D) No potential function exits.
- (E) $f(x,y,z) = 3x^2y + 2y^4z^2 + z^3 + 4z$
- (F) $f(x,y,z) = 3x^2y + 5$
- **(G)** (filled) $f(x,y,z) = 3x^2y + 2y^4z^2 + z^3 + 4$
- (H) $f(x,y,z) = 3x^2y + 2y^4z^2 + 2$

$f(x,y,z) = \int 6xy\,dx = 3x^2y + g_1(y,z)$

$f(x,y,z) = \int (3x^2+8y^3z^2)\,dy = 3x^2y + 2y^4z^2 + g_2(x,z)$

$f(x,y,z) = \int (4y^4z + 3z^2)\,dz = 2y^4z^2 + z^3 + g_3(x,y)$

$f(x,y,z) = 3x^2y + 2y^4z^2 + z^3 + C$

can include any constant

(3.5) (4 pts) For each of the following vector fields, the domain of $\mathbf{F}$ is $\mathbb{R}^2$ which is simply connected. Which vector fields are conservative? **Select <u>all</u> that apply**. No credit will be given if all answers are left blank or all answers are selected.

check if $\frac{\partial F_2}{\partial x} = \frac{\partial F_1}{\partial y}$

- [ ] (a) $\mathbf{F} = \langle 4x^2 - 4y^2, 8xy - 8y\rangle$ $8y \neq -8y$
- [ ] (b) $\mathbf{F} = \langle 3x^2 - 4y, 4y^2 - 2x\rangle$ $-2 \neq -4$
- [x] (c) $\mathbf{F} = \langle 2xe^{xy} + x^2ye^{xy}, x^3e^{xy} + 2y\rangle$
- [ ] (d) $\mathbf{F} = \langle 2y^2, (x+2y)^2\rangle$ $2(x+2y) \neq 4y$
- [x] (e) $\mathbf{F} = \langle e^y, xe^y + y\rangle$ $e^y = e^y$
- [ ] (f) $\mathbf{F} = \langle x^2 - yx, y^2 - xy\rangle$ $-y \neq -x$
- [ ] (g) $\mathbf{F} = \langle xy, x^2 + x - y\rangle$ $2x+1 \neq x$
- [x] (h) $\mathbf{F} = \langle 6\cos(y), -6x\sin(y)\rangle$ $-6\sin(y) = -6\sin(y)$ ✓

(c) $3x^2e^{xy} + x^3ye^{xy} \overset{\checkmark}{=} 2x^2e^{xy} + x^3ye^{xy} + x^2e^{xy}$

4. (5 pts) For each true/false question, fill in the appropriate bubble. No work required.

(4.1) (T) **(F)** If $\mathbf{F}$ is a vector field, then $\text{div}(\mathbf{F})$ is a vector field.

(4.2) (T) **(F)** $\int_0^1 \int_0^z \int_{x+y}^{2x+y} e^{x+y+z}\,dz\,dy\,dx$ is a meaningful triple integral.

(4.3) **(T)** (F) For a differentiable function f, ∇f_P points in the direction of steepest ascent at the point P.

(4.4) **(T)** (F) If $\mathcal{D}$ is closed and bounded, then $f(x,y)$ takes on both an absolute maximum and absolute minimum value on $\mathcal{D}$.

(4.5) (T) **(F)** The integral $\int_{-1}^2 \int_0^x x^2y\,dy\,dx = \left(\int_{-1}^2 x^2\,dx\right)\left(\int_0^x y\,dy\right)$

5. (6 pts) Let $\mathcal{B} = [-1,2] \times [0,4] \times [1,3]$. Fill in the blanks to compute the triple integral over $\mathcal{B}$ **in the order indicated**.

$$\iiint_{\mathcal{B}} f(x,y,z)\,dV = \int_{\boxed{A}}^{\boxed{B}} \int_{\boxed{C}}^{\boxed{D}} \int_{\boxed{E}}^{\boxed{F}} f(x,y,z)\,dz\,dx\,dy$$

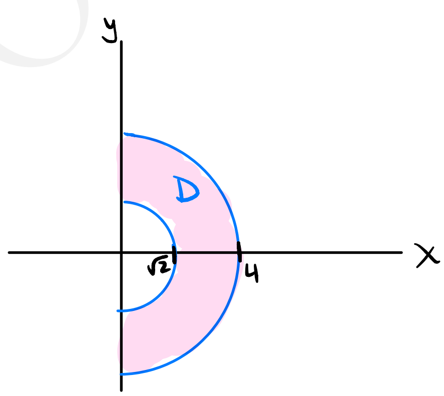
$\boxed{A} =$ 0 $\boxed{B} =$ 4 $\boxed{C} =$ -1 $\boxed{D} =$ 2 $\boxed{E} =$ 1 $\boxed{F} =$ 3

6. (10 pts) Let $\mathcal{D}$ be the region bounded by the circles $x^2+y^2=2$ and $x^2+y^2=16$ in the right half-plane, $x \geq 0$. Fill in the blanks to set up the integral in polar coordinates. **Do not compute the integral.** Hint: Sketching the region of integration might be helpful.

$$\iint_{\mathcal{D}} 3xy^2\,dA = \int_{\boxed{A}}^{\boxed{B}} \int_{\boxed{C}}^{\boxed{D}} \boxed{E}\,dr\,d\theta$$

$\boxed{A} =$ $-\pi/2$ $\boxed{B} =$ $\pi/2$ $\boxed{C} =$ $\sqrt{2}$ $\boxed{D} =$ 4

$\boxed{E} =$ $3r^4\cos\theta\sin^2\theta$



$D: -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$

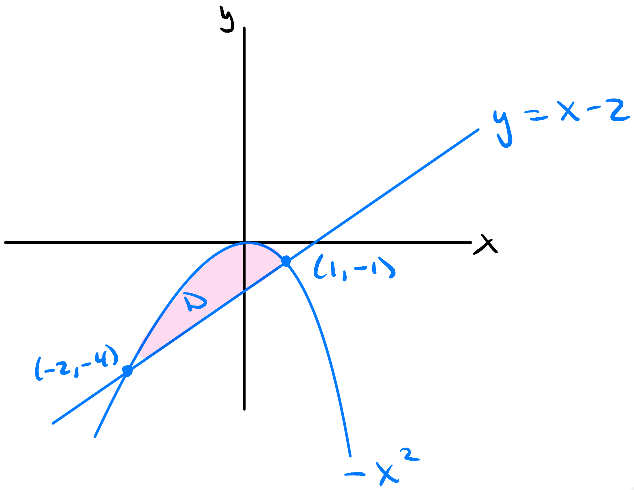
$\sqrt{2} \leq r \leq 4$

$$\iint_D 3xy^2\,dA = \int_{\sqrt{2}}^{4} \int_{-\pi/2}^{\pi/2} \underbrace{3r\cos\theta\, r^2\sin^2\theta\, r}_{3r^4\cos\theta\sin^2\theta}\,dr\,d\theta$$

7. (12 pts) Evaluate $\iint_D 4x\, dA$, where D is the region bounded by $y = -x^2$ and $y = x - 2$. Show all work and write only your final simplified answer in the space provided. Hint: Sketching the region of integration might be helpful.

Answer: -9

Show work here:



Points of intersection:

$-x^2 = x - 2$

$\Rightarrow x^2 + x - 2 = 0$

$\Rightarrow (x+2)(x-1) = 0$

$\Rightarrow x = -2, x = 1$

$(y = -4)$ $(y = -1)$

$D: -2 \le x \le 1$
$x - 2 \le y \le -x^2$ (vertically simple)

$$\iint_D 4x\, dA = \int_{-2}^{1}\int_{x-2}^{-x^2} 4x\, dy\, dx$$

$$= \int_{-2}^{1} 4xy \Big|_{y=x-2}^{y=-x^2} dx$$

$$= \int_{-2}^{1} \left(4x(-x^2) - 4x(x-2)\right) dx$$

$$= \int_{-2}^{1} (-4x^3 - 4x^2 + 8x)\, dx$$

$$= \left(-x^4 - \frac{4}{3}x^3 + 4x^2\right)\Big|_{x=-2}^{x=1}$$

$$= \left(-1 - \frac{4}{3} + 4\right) - \left(\cancel{-16} + \frac{32}{3} + \cancel{16}\right)$$

$$= -9$$

8. (12 pts) Let $\mathcal{C}$ be the curve parameterized by $\mathbf{r}(t) = \langle e^t, 2t^2, 4t \rangle$ for $0 \le t \le 1$. Complete the set up for the line integrals by filling in the blanks with the **simplified** integrands. Show all work. Do **NOT** compute the integral.

(8.1) $\int_{\mathcal{C}} f(x,y,z)\,ds$, where $f(x,y,z) = x^2 + 4z$

$$\int_{\mathcal{C}} f(x,y,z)\,ds = \int_0^1 \underline{(e^{2t}+16t)\sqrt{e^{2t}+16t^2+16}}\,dt$$

(8.2) $\int_{\mathcal{C}} \mathbf{F}\cdot d\mathbf{r}$, where $\mathbf{F} = \left\langle \frac{7y}{x}, -z, 3x \right\rangle$

$$\int_{\mathcal{C}} \mathbf{F}\cdot d\mathbf{r} = \int_0^1 \underline{(-2t^2+12e^t)}\,dt$$

Show all work needed to justify your answers here:

$$\vec{r}\,'(t) = \langle e^t, 4t, 4 \rangle$$

$$\|\vec{r}\,'(t)\| = \sqrt{e^{2t} + 16t^2 + 16}$$

$$f(\vec{r}(t)) = e^{2t} + 16t$$

$$\vec{F}(\vec{r}(t)) = \left\langle \frac{14t^2}{e^t}, -4t, 3e^t \right\rangle$$

$$\vec{F}(\vec{r}(t))\cdot\vec{r}\,'(t) = \left\langle \frac{14t^2}{e^t}, -4t, 3e^t \right\rangle \cdot \langle e^t, 4t, 4 \rangle$$

$$= \frac{14t^2}{\cancel{e^t}}\cancel{e^t} - 16t^2 + 12e^t$$

$$= -2t^2 + 12e^t$$

9. (10 pts) Let $f(x,y,z) = 9x^2y - 8z$ be a potential function for the conservative vector field $\mathbf{F}$.

(9.1) Calculate $\int_C \mathbf{F} \cdot d\mathbf{r}$ for the path $\mathbf{r}(t) = \langle t, t^2, 2\rangle$, $0 \le t \le 1$. Show all work and write only your simplified final answer in the box provided.

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \boxed{9}$$

Show work here:

$P = \vec{r}(0) = (0,0,2)$

$Q = \vec{r}(1) = (1,1,2)$

$$\int_C \vec{F} \cdot d\vec{r} = f(Q) - f(P) = (9(1)^2(1) - 8(2)) - (9(0)^2(0) - 8(2)) = 9$$

Could also do:

$\vec{F} = \nabla f = \langle 18xy, 9x^2, -8\rangle$, $\vec{r}\,'(t) = \langle 1, 2t, 0\rangle$

$\vec{F}(\vec{r}(t)) = \langle 18t^3, 9t^2, -8\rangle$

$\vec{F}(\vec{r}(t)) \cdot \vec{r}\,'(t) = 18t^3 + 18t^3 + 0 = 36t^3$

$$\int_C \vec{F} \cdot d\vec{r} = \int_0^1 36t^3\,dt = 9t^4\Big|_0^1 = 9$$

(9.2) **T** F The value of $\int_C \mathbf{F} \cdot d\mathbf{r}$ will be the same as 9.1 using the path $\mathbf{r_2}(t) = \langle t, t, 2\rangle$, $0 \le t \le 1$.

Justify your answer:

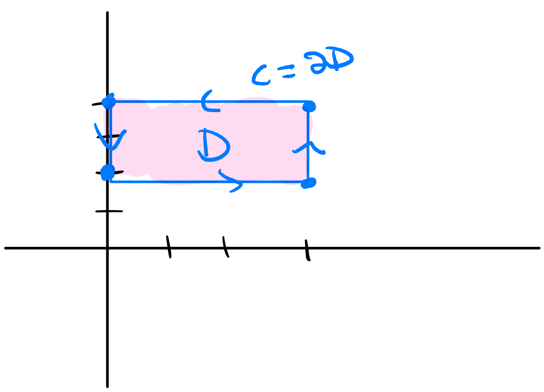
$P = \vec{r}_2(0) = (0,0,2)$ and $Q = \vec{r}_2(1) = (1,1,2)$

Since $\vec{F}$ is conservative, it is path independent so $\int_C \vec{F} \cdot d\vec{r}$ will be the same for any path from P to Q

10. (10 pts) **Use Green's Theorem** to evaluate the line integral $\oint_{\mathcal{C}} (x+2y^2)\,dx + (7y+x^2)\,dy$, where $\mathcal{C}$ is the boundary of the rectangle with vertices $(0,2)$, $(3,2)$, $(3,4)$, and $(0,4)$ traversed counterclockwise. Show all your work and write only the final simplified answer in the box provided.

Answer: $\boxed{-54}$

Show work here:



$\vec{F} = \langle F_1, F_2 \rangle = \langle x+2y^2, 7y+x^2 \rangle$

$\dfrac{\partial F_2}{\partial x} - \dfrac{\partial F_1}{\partial y} = 2x - 4y$

$D: \; 0 \le x \le 3$
$\quad\;\; 2 \le y \le 4$

$$\oint_C (x+2y^2)\,dx + (7y+x^2)\,dy = \int_2^4 \int_0^3 (2x-4y)\,dx\,dy$$

$$= \int_2^4 (x^2 - 4xy)\Big|_{x=0}^{x=3} dy$$

$$= \int_2^4 (9 - 12y)\,dy$$

$$= (9y - 6y^2)\Big|_{y=2}^{y=4}$$

$$= (9(4) - 6(4)^2) - (9(2) - 6(2)^2)$$

$$= -54$$